

Joel Teply

Principal / Staff Software Engineer

Distributed Systems · Cloud · Backend · AI/ML · High-Performance Computing

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Profile

Full-stack software engineer and technical founder with 25+ years building production systems from databases, backend services, and web applications through embedded devices, mobile SDKs, GPU computing, cloud platforms, computer vision, and modern distributed AI.

I work across layers rather than treating them as separate disciplines: architecture, data, APIs, concurrency, infrastructure, performance, security, and the product above them. Long track record taking difficult systems from prototype through production, supporting enterprise customers, and making software reliable on constrained hardware and at scale.

Experience

Cambrian Tech — Co-founder, CTO, Principal Engineer · 2009–present

Co-founded and technically led a computer vision and software company building proprietary technology and custom engineering programs for enterprise customers across home improvement, retail, and automotive. Owned architecture and implementation across backend/cloud systems, ML, real-time C++, GPU computing, native/mobile SDKs, web applications, data pipelines, and customer integrations. Technology was licensed to Fortune 500 customers including Home Depot and Lowe's and deployed to millions of users.

Cloud, backend & data. Built production Python/TensorFlow processing services and elastically scaling AWS infrastructure for image-analysis products, including APIs, S3-backed data flow, Docker CPU/GPU deployment, automated tests, model utilities, and production monitoring. Built backend and web systems across TypeScript/Node.js and other stacks, REST and GraphQL APIs, SQL and NoSQL data stores, serverless services, CloudFormation, and customer integration infrastructure.

Systems & platform engineering. Built a cross-platform C++ runtime with priority scheduling, backpressure, lazy shared data representations, SIMD, threading, and GPU/GPGPU acceleration. The same core architecture supported iOS, Android/JNI, Unity, browser, and headless/server

targets. Designed SDKs to keep customer-facing interfaces thin while isolating complex native, ML, and platform-specific internals.

Technical leadership & customers. Served as Cambrian's technical lead on most client-facing calls, working with customers from technical discovery and feasibility through architecture, integration, and delivery. Participated in sales calls, product demonstrations, trade shows, technical/legal discussions, and investor and competition pitches. Cambrian won the inaugural \$100,000 LaunchKC Grand Prize and was a top-five finalist at Atlanta Startup Battle 6.0.

Applied ML & perception. Developed and productionized computer-vision and ML systems including segmentation, tracking, 3D scene understanding, model training, synthetic data, quantization/compression, and real-time on-device inference. First-named inventor on US10964097B2.

TripleBlind → Ideem — Software Engineer, Cybersecurity · 2023–2026

Built enterprise authentication and SDK systems involving device-bound identity and key handling, PKI/encryption, secure client integration, biometrics/platform security, and cross-platform APIs.

EyeVerify / ZOLOZ — Software Engineer, Computer Vision & Biometrics · 2012–2016

One of the first two engineers turning biometric research into a commercial SDK on commodity phones. Built real-time image-processing and authentication systems spanning native camera pipelines, tracking, segmentation, multi-frame fusion, OpenGL/GLSL acceleration, concurrency, and anti-spoofing/liveness. Named on three patents; stayed through acquisition and integration.

Propaganda3 — Software Engineer · 2010–2012

Primarily mobile and product engineering for agency clients. Built H&R Block's first tax app in roughly three months, including OCR/photo W-2 ingestion and a reusable engine supporting federal and state tax forms; the architecture was reused for the Android implementation. Also shipped iOS fitness, media, and other consumer applications.

VML — Software Engineer · 2006–2010

Full-stack agency engineering for major brands across C#/.NET and CMS-backed web systems, frontend applications, databases/SQL, Java, backend integrations, and customer-facing digital products.

Earlier · 1999–2006

Full-stack and systems engineering across AllofE Solutions, Federal Home Loan Bank of Topeka, NovaStar Financial, independent contracting, and AgileWise. Work included PHP, Perl, Java, LAMP/MySQL, ASP.NET/C#, SQL-heavy enterprise applications, custom ORMs and web servers, PCB design, embedded hardware, RFID, touchscreen systems, Windows CE/Linux bring-up, and display drivers.

Current Systems Work

Continuum — large Rust platform for persistent AI agents across commodity hardware.

Architecture spans distributed compute, P2P communication, scheduling and resource governance, inference-runtime internals, KV-cache/slot management, Metal/CUDA placement, telemetry, evaluation, storage, and multiple concurrent AI workloads. Maintains a llama.cpp fork with substantial runtime work.

github.com/CambrianTech/continuum

AIRC — Rust peer-to-peer communication substrate connecting AI agents across tabs, machines, and accounts with end-to-end encrypted messaging and file sharing; foundation for Continuum's distributed communication layer.

github.com/CambrianTech/airc

Sentinel-AI — Python/PyTorch transformer architecture and model-efficiency research spanning LoRA, mixed precision, structured pruning, quantization, evaluation, and reproducible model forging.

github.com/CambrianTech/sentinel-ai

Technical

Languages — Python, Java, C#, C, C++, Rust, Go, TypeScript/JavaScript, Objective-C/Objective-C++, Swift, Kotlin, PHP, Perl, Lua, shell/Bash, Visual Basic, assembly; historical ASP/ASP.NET, ColdFusion and Java applets

Backend, cloud & data — extensive AWS, S3, CloudFormation, serverless systems, Docker, Node.js/TypeScript, .NET, REST, GraphQL, message-driven systems, data/ML pipelines, SQL, SQL Server, MySQL, PostgreSQL, SQLite, NoSQL/Dgraph, relational data modeling, custom ORMs and web servers

Architecture & systems — distributed systems, API/SDK design, concurrency, scheduling and backpressure, performance optimization, testing, telemetry/observability, memory-constrained execution, Linux, embedded and edge systems

GPU & high-performance computing — SIMD, Metal, CUDA, OpenGL/GLSL, OpenGL ES GPGPU, WebGL, shaders, native camera/media pipelines

Web & product — React, Node.js, TypeScript/JavaScript, HTML/CSS/SCSS, CMS platforms, frontend/backend web systems, browser graphics/ML, mobile and web SDKs

Security — PKI, OpenSSL, encryption, device identity, secure enclaves, Face ID/biometric integration, device-bound authentication and cryptographic client integration

AI/ML — PyTorch, TensorFlow, computer vision, model training/evaluation, synthetic data, CNNs, transformers/LLMs, agents, RAG, LoRA, quantization, structured pruning, inference runtimes

Education & Patents

B.S. Computer Engineering — University of Kansas

- **US10964097B2** — segmentation systems and methods; first-named inventor
- **US9665784B2, US8437513B1, US8675925B2** — spoof detection / liveness
- \$100,000 LaunchKC Grand Prize winner; Atlanta Startup Battle 6.0 top-five finalist